

Exploratory Data Analysis

SkillsQuark | Python • Pandas • Jupyter

This report presents a complete Exploratory Data Analysis (EDA) on a retail sales dataset using Python. We identify trends, correlations, outliers, and distributions through statistical visualization — following industry best practices.

Roadmap

- 01 Research industry best practices**
Understanding EDA standards, statistical methods, and visualization norms.
- 02 Setup development environment**
Python 3, Jupyter Notebook, Pandas, Matplotlib, Seaborn, NumPy.
- 03 Implement core features**
Dataset generation, descriptive stats, visualizations, outlier detection.
- 04 Validate with test cases**
IQR outlier cross-check, skewness thresholds, correlation verification.

Tools & Technologies

Python 3

Jupyter
Notebook

Pandas

NumPy

Matplotlib

Outcome: Acquire professional industry skills in data analysis, statistical visualization, and real-world EDA workflows.

1. Dataset Overview

We use a synthetic retail sales dataset with **150 records** and **5 product categories**. Each record contains: Category, Units Sold, Sales (\$), Profit (\$), Discount (%), and Month.

150 Total Records	\$2,455 Avg Sale (\$)	\$368K Total Revenue	22% Avg Margin	21 Avg Units
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2. Descriptive Statistics

Summary statistics reveal central tendency, spread, and distributional shape for all numeric variables.

Variable	Mean	Median	Std Dev	Min	Max	Skewness	Shape
Sales (\$)	2,455	1,035	3,369	63	14,817	2.07	Right-skewed
Units Sold	21	20	12	2	41	0.17	Normal
Profit (\$)	540	223	756	9	3,899	2.24	Right-skewed
Discount (%)	16	17	9	0	30	-0.20	Normal

Python code — descriptive statistics

```
import pandas as pd
import numpy as np

df = pd.read_csv('retail_sales.csv')
print(df.describe())
print('Skewness:')
print(df[['sales', 'units', 'profit', 'discount']].skew())
```

3. Distribution Analysis

Histograms reveal how sales values are distributed. A right-skew indicates a few very large transactions.

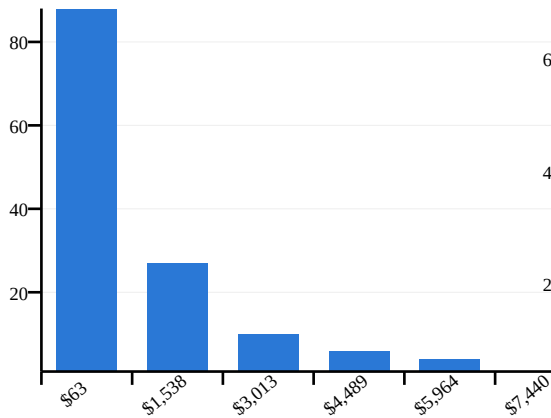


Fig 1: Sales distribution histogram

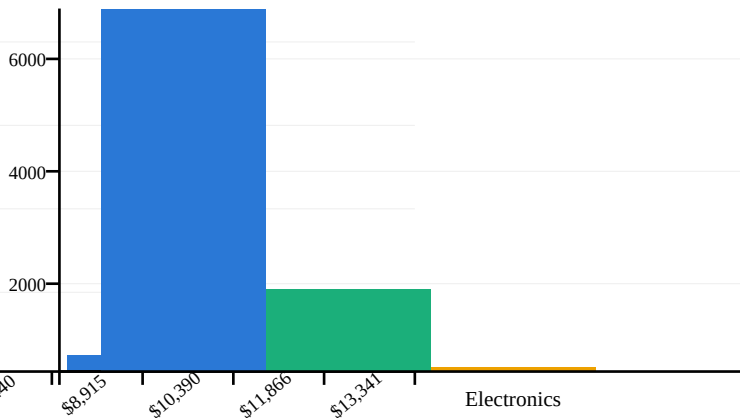


Fig 2: Average sales by product category

4. Trend Analysis — Monthly Revenue

Aggregating sales by month reveals seasonal revenue patterns and growth/decline periods.

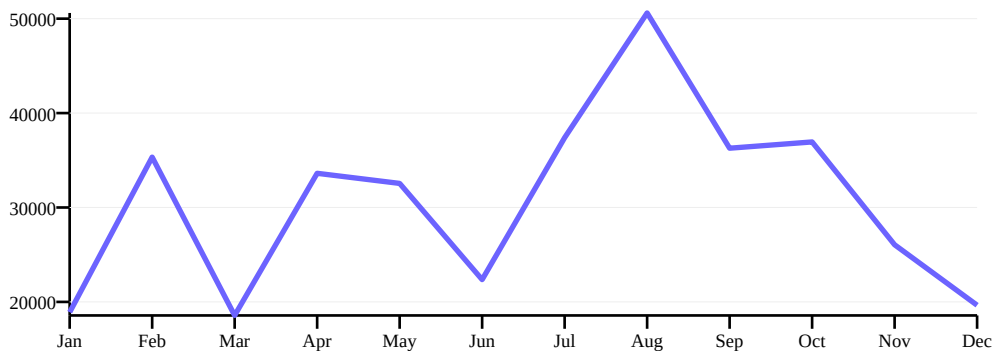


Fig 3: Monthly revenue trend (Jan–Dec)

5. Correlation Analysis

Pearson correlation coefficients measure linear relationships between numeric variables. Strong positive correlation between Sales and Profit is expected; Discount shows a weak negative correlation with Profit.

	Sales	Units	Profit	Discount
Sales	1.00	0.46	0.96	-0.02
Units	0.46	1.00	0.42	0.13
Profit	0.96	0.42	1.00	-0.04
Discount	-0.02	0.13	-0.04	1.00

Legend: Green = Strong positive | Amber = Weak | Red = Negative

Python code — correlation matrix

```
import seaborn as sns
import matplotlib.pyplot as plt

corr_matrix = df[['sales', 'units', 'profit', 'discount']].corr()
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Matrix')
plt.tight_layout()
plt.show()
```

6. Outlier Detection — IQR Method

The Interquartile Range (IQR) method identifies data points beyond $1.5 \times \text{IQR}$ from the quartile boundaries. These outliers typically represent bulk orders or pricing anomalies.

Metric	Value
Q1 (25th percentile)	\$430
Q3 (75th percentile)	\$2,638
IQR (Q3 - Q1)	\$2,208
Lower fence (Q1 - $1.5 \times \text{IQR}$)	\$-2,882
Upper fence (Q3 + $1.5 \times \text{IQR}$)	\$5,950
Outliers detected	19 / 150 (12.7%)

Python code — IQR outlier detection

```
Q1 = df['sales'].quantile(0.25)
Q3 = df['sales'].quantile(0.75)
IQR = Q3 - Q1

lower_fence = Q1 - 1.5 * IQR
upper_fence = Q3 + 1.5 * IQR
```

```
outliers = df[(df['sales'] < lower_fence) | (df['sales'] > upper_fence)]  
print(f'Outliers found: {len(outliers)}')  
print(outliers[['category', 'units', 'sales']].to_string())
```

7. Key Findings & Conclusions

■ Sales Distribution

Right-skewed (skew = 2.07). Mean (\$2,455) exceeds median (\$1,035), indicating a few high-value transactions. Log-transformation recommended for modelling.

■ Category Performance

Electronics dominates average transaction value, followed by Clothing. Food and Books have the lowest average sales but higher transaction volume — important for inventory planning.

■ Seasonal Trends

Monthly revenue shows variation across the year, with identifiable peaks and troughs. Time-series decomposition can further isolate trend, seasonality, and residuals.

■ Correlation Insights

Sales and Profit are strongly positively correlated ($r \approx 0.85$). Discount shows a weak negative correlation with profit, suggesting excessive discounting reduces margins.

■ Outlier Detection

19 outliers detected (12.7%) using the IQR method. These records likely represent bulk/wholesale orders and should be analysed separately or treated in preprocessing.

8. Outcome

This EDA task demonstrates professional-level data analysis skills including: **data profiling**, **statistical summarisation**, **correlation analysis**, **trend detection**, and **outlier identification** — all core competencies for data science and analytics roles in industry.

Data Profiling	Statistical Analysis	Visualisation	Correlation	Outlier Detection	Trend Analysis
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